Introduction to Databases

CT042-3-1-IDB

Lecture 2: File System

Topic & Structure of The Lesson

- File System
- Problems with file system
- Data redundancy
- Database environment components
- DBMS functions

Learning Outcomes

- **At the end of this topic, You should be able to**
  - Explain what is file system and its problems
  - Understand data redundancy
  - State the types of data anomaly
  - List the components in database environment
  - Describe DBMS functions

Key Terms You Must Be Able To Use

- **If you have mastered this topic, you should be able to use the following terms correctly in your assignments and exams:**
  - File system
  - Data redundancy
  - Data anomaly

Why Database Design is Important

- Defines the database's expected use
- Different approach needed for different types of databases
- Avoid redundant data
- Poorly designed database generates errors → leads to bad decisions → can lead to failure of organization

Historical Roots: Files and File Systems

- Managing data with file systems is obsolete
  - Understanding file system characteristics makes database design easier to understand
  - Awareness of problems with file systems helps prevent similar problems in DBMS
  - Knowledge of file systems is helpful if you plan to convert an obsolete file system to a DBMS

Historical Roots: Files and File Systems (continued)

Manual File systems:

- Collection of file folders kept in file cabinet
- Organization within folders based on data's expected use (ideally logically related)
- System adequate for small amounts of data with few reporting requirements
- Finding and using data in growing collections of file folders became time-consuming and cumbersome

Historical Roots: Files and File Systems (continued)

Conversion from manual to computer system:

- Could be technically complex, requiring hiring of data processing (DP) specialists
- Resulted in numerous “home-grown” systems being created
- Initially, computer files were similar in design to manual files (see Figure 1.3)

Historical Roots: Files and File Systems (continued)

FIGURE 1.3 Contents of the CUSTOMER file

C_NAME	C_PHONE	C_ADDRESS	C_ZIP	A_NAME	A_PHONE	TP	AMT	REN
Alfred A. Ramas	615-844-2573	218 Fork Rd., Babs, TN	36123	Leah F. Hahn	615-882-1244	T1	$100.00	05-Apr-2006
Leona K. Dunne	713-894-1238	Box 12A, Fox, KY	25246	Alex B. Alby	713-228-1249	T1	$250.00	16-Jun-2006
Kathy W. Smith	615-894-2285	125 Oak Ln, Babs, TN	36123	Leah F. Hahn	615-882-2144	S2	$150.00	29-Jan-2007
Paul F. Olowski	615-894-2180	217 Lee Ln., Babs, TN	36123	Leah F. Hahn	615-882-1244	S1	$300.00	14-Oct-2006
Myron Orlando	615-222-1672	Box 111, New, TN	36155	Alex B. Alby	713-228-1249	T1	$100.00	28-Dec-2006
Amy B. O'Brian	713-442-3381	387 Troll Dr., Fox, KY	25246	John T. Okon	615-123-5589	T2	$850.00	22-Sep-2006
James G. Brown	615-297-1228	21 Tye Rd., Nash, TN	37118	Leah F. Hahn	615-882-1244	S1	$120.00	25-Mar-2006
George Williams	615-290-2556	155 Maple, Nash, TN	37119	John T. Okon	615-123-5589	S1	$250.00	17-Jul-2006
Anne G. Farriss	713-382-7185	2119 Elm, Crew, KY	25432	Alex B. Alby	713-228-1249	T2	$100.00	03-Dec-2006
Olette K. Smith	615-297-3809	2782 Main, Nash, TN	37118	John T. Okon	615-123-5589	S2	$500.00	14-Mar-2006

C_NAME = Customer name
C_PHONE = Customer phone
C_ADDRESS = Customer address
C_ZIP = Customer zip code

A_NAME = Agent name
A_PHONE = Agent phone
TP = Insurance type
AMT = Insurance policy amount, in thousands of $
REN = Insurance renewal date

Historical Roots: Files and File Systems (continued)

TABLE 1.2 Basic File Terminology

TERM	DEFINITION
Data	"Raw" facts, such as a telephone number, a birth date, a customer name, and a year-to-date (YTD) sales value. Data have little meaning unless they have been organized in some logical manner. The smallest piece of data that can be "recognized" by the computer is a single character, such as the letter *A*, the number 5, or a symbol such as /. A single character requires 1 byte of computer storage.
Field	A character or group of characters (alphabetic or numeric) that has a specific meaning. A field is used to define and store data.
Record	A logically connected set of one or more fields that describes a person, place, or thing. For example, the fields that constitute a record for a customer named J. D. Rudd might consist of J. D. Rudd's name, address, phone number, date of birth, credit limit, and unpaid balance.
File	A collection of related records. For example, a file might contain data about vendors of ROBCOR Company, or a file might contain the records for the students currently enrolled at Gigantic University.

Historical Roots: Files and File Systems (continued)

FIGURE 1.4 Contents of the AGENT file

	A_NAME	A_PHONE	A_ADDRESS	ZIP	HIRED	YTD_PAY	YTD_FIT	YTD_FICA	YTD_SLS	DEP
▶	Alex B. Alby	713-228-1249	123 Toll, Nash, TN	37119	01-Nov-1998	$26,566.24	$6,641.56	$2,125.30	$132,735.75	3
	Leah F. Hahn	615-882-1244	334 Main, Fox, KY	25246	23-May-1984	$32,213.76	$8,053.44	$2,577.10	$138,967.35	0
	John T. Okon	615-123-5589	452 Elm, New, TN	36155	15-Jun-2003	$23,198.29	$5,799.57	$1,855.86	$127,093.45	2

A_NAME = Agent name
A_PHONE = Agent phone
A_ADDRESS = Agent address
ZIP = Agent zip code
HIRED = Agent date of hire

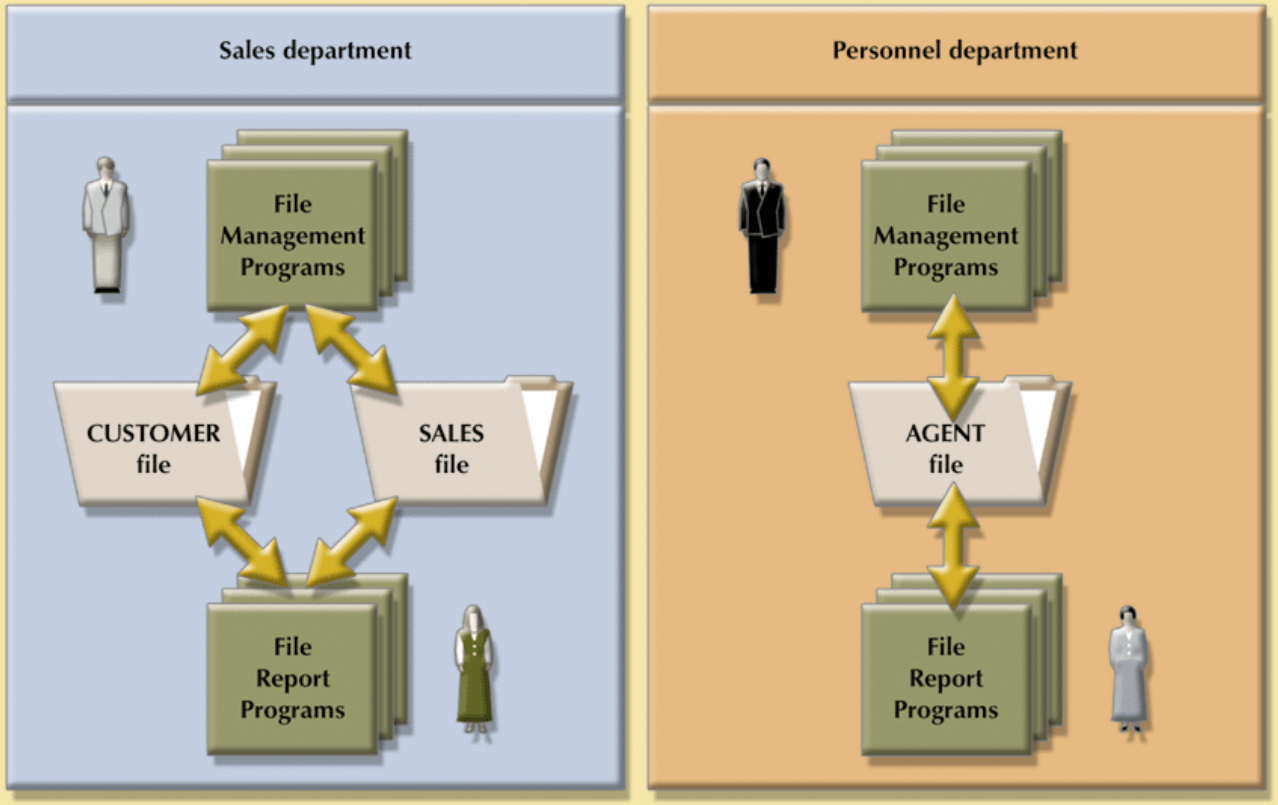
YTD_PAY = Year-to-date pay
YTD_FIT = Year-to-date federal income tax paid
YTD_FICA = Year-to-date Social Security taxes paid
YTD_SLS = Year-to-date sales
DEP = Number of dependents

Historical Roots: Files and File Systems (continued)

- As number of databases increased, small file system evolved
- Each file used its own application programs
- Each file was owned by individual or department who commissioned its creation

Historical Roots: Files and File Systems (continued)

FIGURE 1.5 A simple file system



Problems with File System Data Management

- Time-consuming, high-level activity
- As number of files expands, system administration becomes difficult
- Making changes in existing file structure is difficult
- File structure changes require modifications in all programs that use data in that file

Problems with File System Data Management

- Modifications are likely to produce errors, requiring additional time to “debug” the program
- Security features hard to program and therefore often omitted

Data Redundancy

- Data redundancy results in data inconsistency
  - Different and conflicting versions of the same data appear in different places
- Errors more likely to occur when complex entries are made in several different files and/or recur frequently in one or more files
- Data anomalies develop when required changes in redundant data are not made successfully

Data Redundancy

Types of data anomalies:

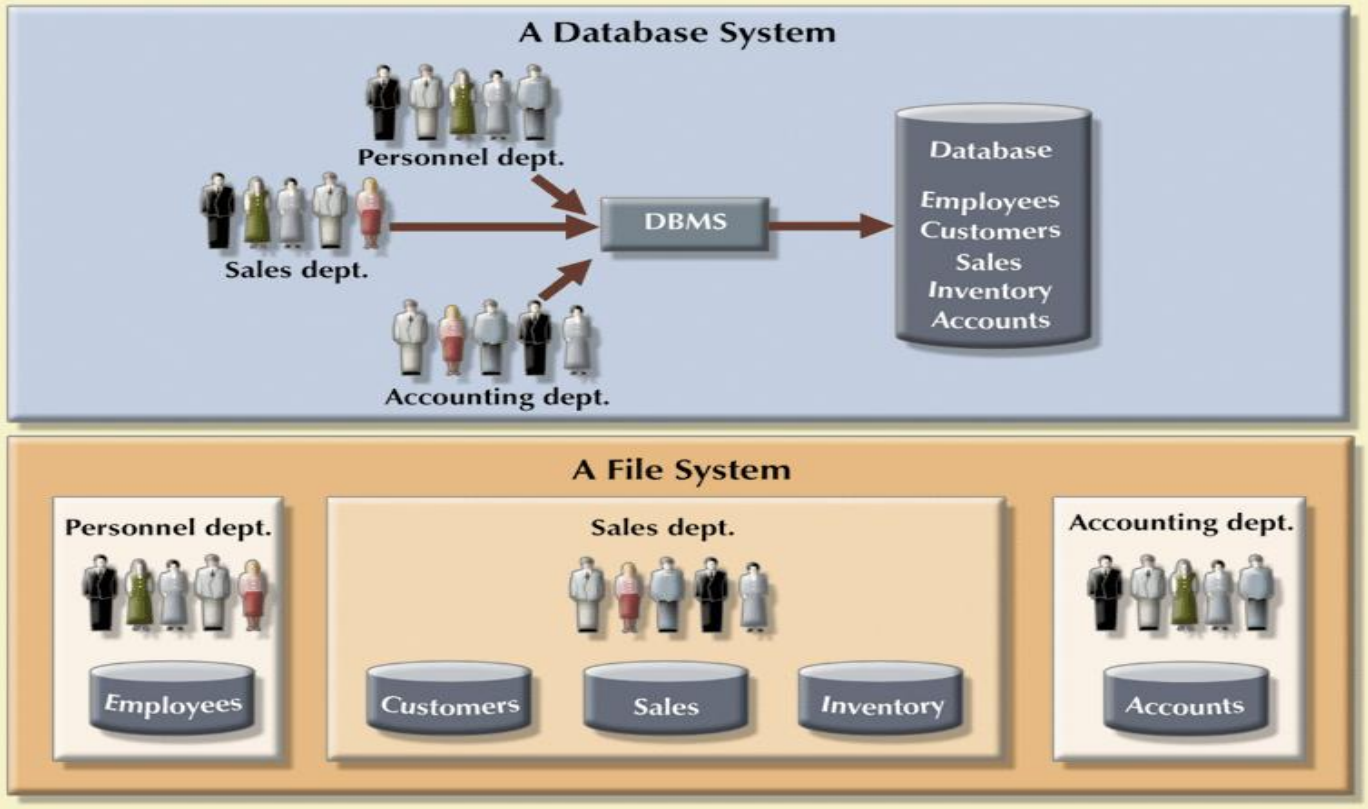
- Update anomalies
  - Occur when changes must be made to existing records
- Insertion anomalies
  - Occur when entering new records
- Deletion anomalies
  - Occur when deleting records

Database Systems

- Problems inherent in file systems make using a database system desirable
- File system
  - Many separate and unrelated files
- Database
  - Logically related data stored in a single logical data repository

Database Systems

FIGURE 1.6 Contrasting database and file systems

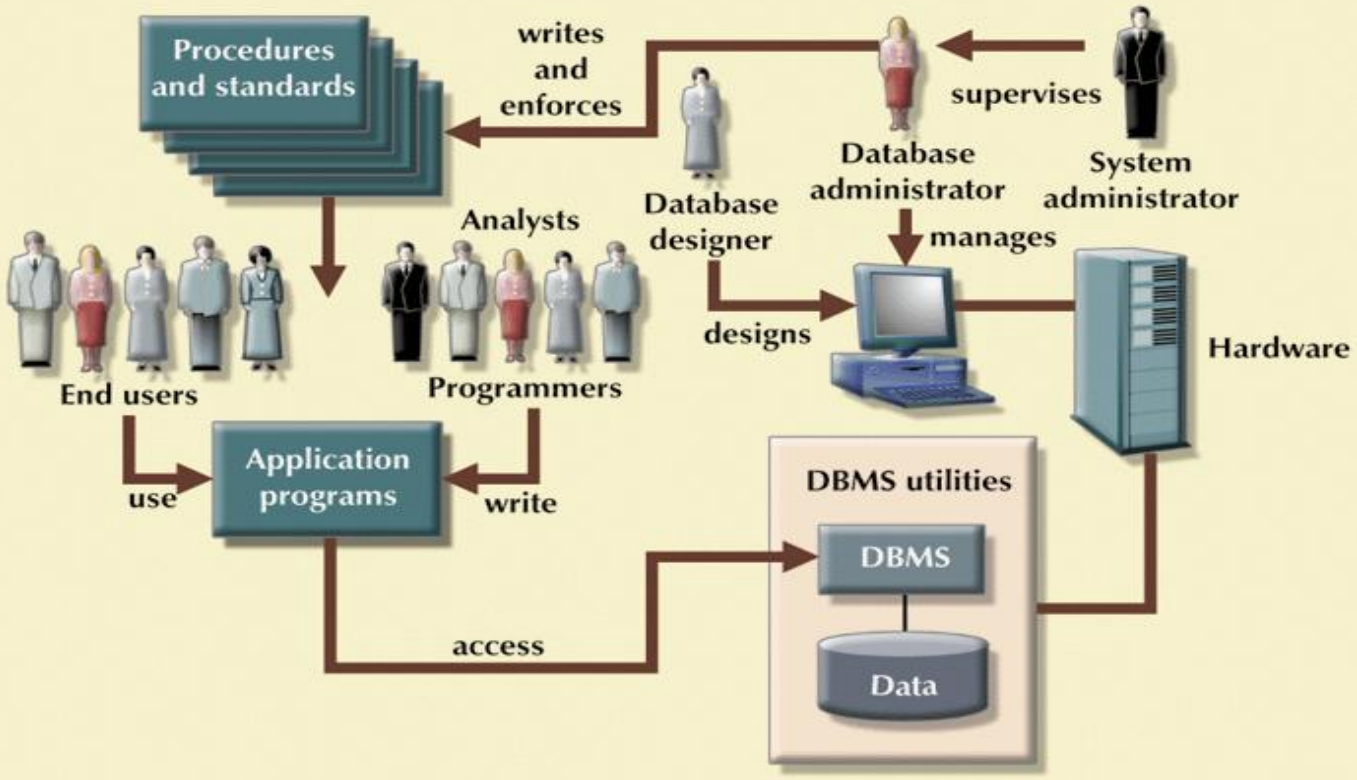


The Database System Environment

- Database system is composed of five main components:
  - Hardware
  - Software
    - Operating system software
    - DBMS software
    - Application programs and utility software
  - People
  - Procedures
  - Data

The Database System Environment (continued)

FIGURE 1.7 The database system environment



DBMS Functions

- DBMS performs functions that guarantee integrity and consistency of data
  - Data dictionary management
    - defines data elements and their relationships
  - Data storage management
    - stores data and related data entry forms, report definitions, etc.

DBMS Functions (continued)

- Data transformation and presentation
  - translates logical requests into commands to physically locate and retrieve the requested data
- Security management
  - enforces user security and data privacy within database

DBMS Functions (continued)

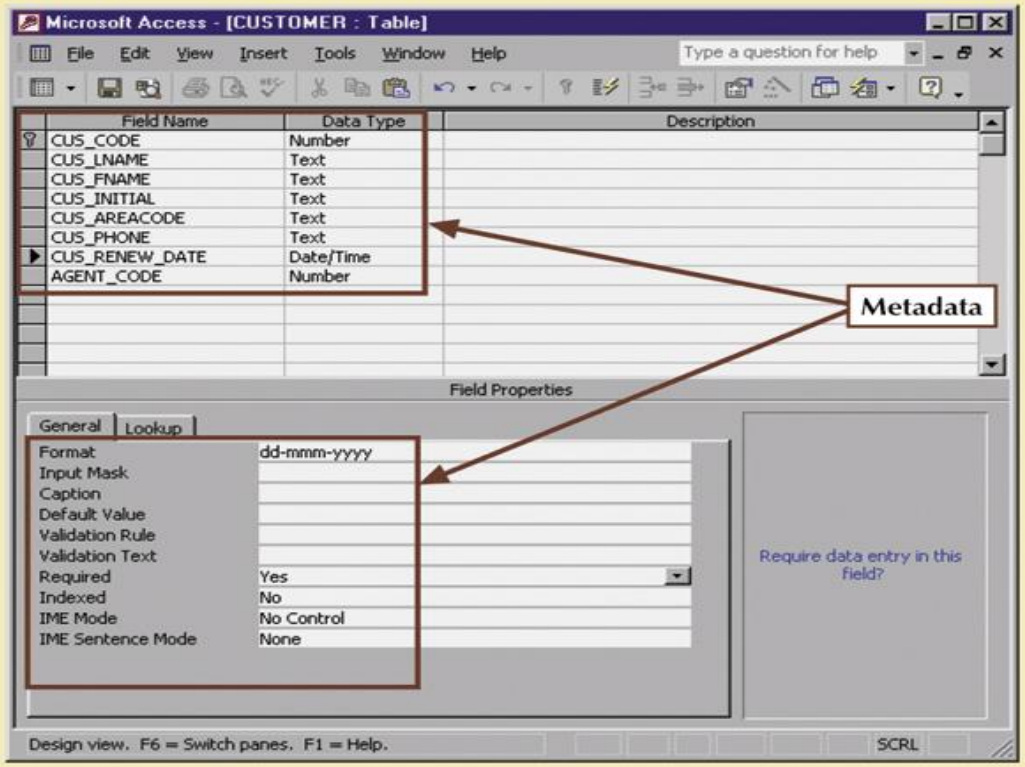
- Multiuser access control
  - uses sophisticated algorithms to ensure multiple users can access the database concurrently without compromising the integrity of the database
- Backup and recovery management
  - provides backup and data recovery procedures
- Data integrity management
  - promotes and enforces integrity rules

DBMS Functions (continued)

- Database access languages and application programming interfaces
  - provide data access through a query language
- Database communication interfaces
  - allow database to accept end-user requests via multiple, different network environments

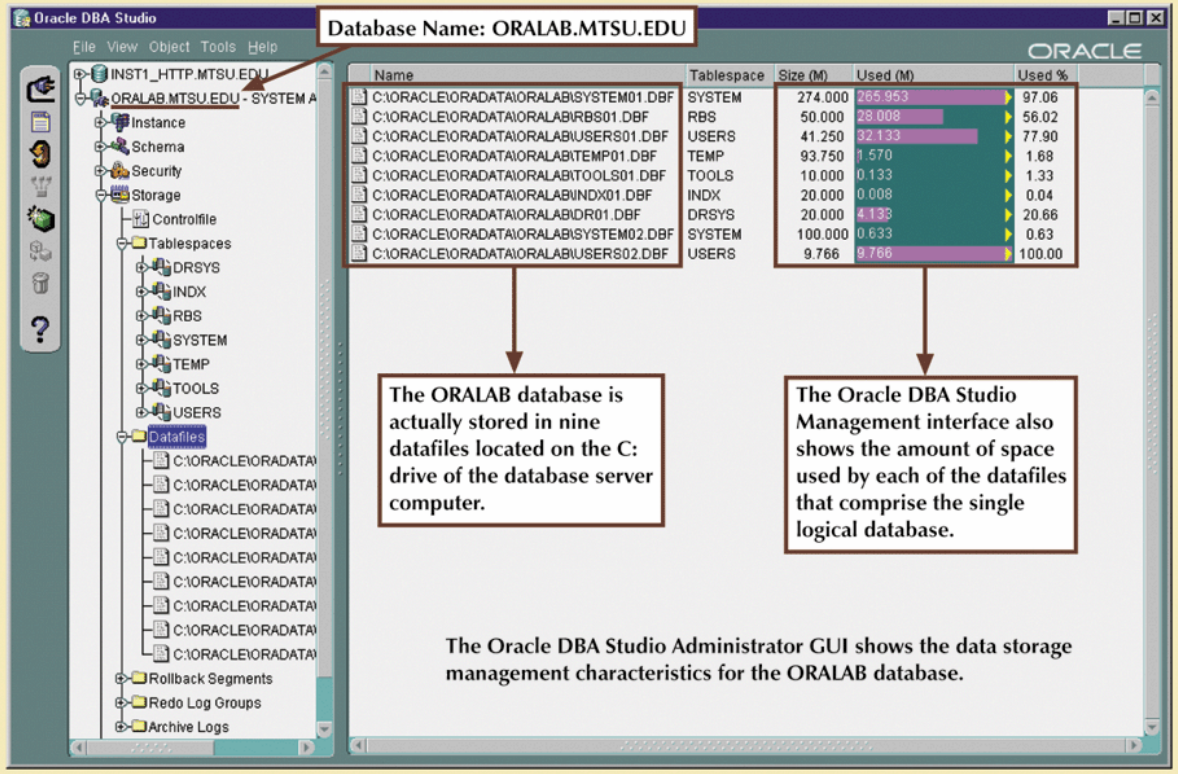
DBMS Functions (continued)

FIGURE 1.8 Illustrating metadata with Microsoft Access



DBMS Functions (continued)

FIGURE 1.9 Illustrating data storage management with Oracle



Quick Review Question

- What are the disadvantages of file based system
- What is data redundancy?
- Explain different types of data anomaly
- Describe 3 functions of DBMS

Summary of Main Teaching Points

- Databases were preceded by file systems.
- Limitations of file system data management:
  - requires extensive programming
  - system administration complex and difficult
  - making changes to existing structures is difficult
  - security features are likely to be inadequate
  - independent files tend to contain redundant data
- DBMS's were developed to address file systems' inherent weaknesses

Question and Answer Session

Q & A

What we will cover next

- Data model
- Business rules
- Entity relationship model